STANDARD COSTING

1. Basic Terms

Revised Bud. on Basis of Actual o/p

Budget	Standard	Actual
↓	↓	↓
Kya Socha tha	Kya Hona Chahiye Tha	Kya Ho Gaya

Variance Analysis

2. Always calculate standard data on the basis of actual output

3. Variances

Variances
- +ve — Favourable = (F)
- −ve — Adverse = (A)

4. Material Variance

Material Cost Variance (MCV) = Std. Mat. Cost − Actual Mat. Cost

Material Price Variance (MPV) = (SP − AP) × AQ

Material Usage Variance (MUV) = (SQ − AQ) × SP

Material Mix Variance (MMV) = (RSQ − AQ) × SP

Material Yield Var. (MYV) = (SQ − RSQ) × SP

(or)

= (AY − SY) × SC P.U. of output

SP = Std. Price
AP = Actual Price
AQ = Actual Qty.
SQ = Std. Qty.
RSQ = Actual I/P in Std. Ratio

AY = Actual O/P
SY = Std. O/P = $\frac{\text{Std. O/P}}{\text{Std. I/P}}$ × Actual I/P

5. Material Cost Variance (MCV)

MCV = Standard cost – Actual Cost

6. Material Price Variance (MPV)

MPV = (SP – AP) × Actual quantity

7. Material Usage Variance (MUV)

MUV = (SQ – AQ) × SP

8. Material Mix Variance (MMV)

MMV = (RSQ – AQ) × SP

9. Material Yield Variance (MYV)

MYV = (SQ – RSQ) × SP

MYV = (AY – SY) × Standard cost per unit of output

SP = Standard price per unit
AP = Actual price per unit
SQ = Standard quantity of material
AQ = Actual quantity of material
RSQ = Revised Standard Quantity
= Actual input in standard ratio
AY = Actual Yield or Output
SY = Standard Yield or Output
$= \frac{Standard\ Output}{Standard\ Input} \times Actual\ Input$

10. Actual Quantity for MPV

Actual Quantity
- Actual quantity purchased
- (Prefer) Actual quantity consumed — opening stock + Purchases - Closing Stock

* If opening stock rate is not given then consider it at standard price
* Unless otherwise provided, FIFO method is used

Question – 1	

The standard cost of a chemical mixture is as follows:

60% of Material A @ ₹ 50 per kg

40% of Material B @ ₹ 60 per kg

A standard loss of 25% on output is expected in production. The cost records for a period has shown the following usage.

540 kg of Material A @ ₹ 60 per kg

260 kg of Material B @ ₹ 50 per kg

The quantity processed was 680 kilograms of good product.

From the above given information, calculate:

(i) Material cost variance

(ii) Material price variance

(iii) Material usage variance
(iv) Material mix variance
(v) Material yield variance

Solution

Basic Calculation

Particulars	Standard			Actual			Revised Std. Quantity
	Quantity	Rate	Amount	Quantity	Rate	Amount	
Material A	850 × 60% = 510	50	25,500	540	60	32,400	800 × 60% = 480
Material B	850 × 40% = 340	60	20,400	260	50	13,000	800 × 40% = 320
Input	850		45,900	800		45,400	800
(-) Loss	680 × 25% = 170			120			
Output	680			680			

Calculation of Variances

(i) Material Cost Variance = Standard Cost – Actual cost

A = 25,500 – 32,400 = ₹ 6,900 (A)
B = 20,400 – 13,000 = ₹ 7,400 (F)
MCV = ₹ 500 (F)

1. Material Price Variance = (SP – AP) × AQ

A = (50 – 60) × 540 = ₹ 5,400 (A)
B = (60 – 50) × 260 = ₹ 2,600 (F)
MPV = ₹ 2,800 (A)

2. Material Usage (or Quantity) Variance = (SQ – AQ) × SP

A = (510 – 540) × 50 = ₹ 1,500 (A)
B = (340 – 260) × 60 = ₹ 4,800 (F)
MUV = ₹ 3,300 (F)

3. Material Mix Variance = (RSQ - AQ) × SP

A = (480 – 540) × 50 = ₹ 3,000 (A)
B = (320 – 260) × 60 = ₹ 3,600 (F)
MMV = ₹ 600 (F)

4. Material Yield Variance = (SQ - RSQ) × SP

A = (510 – 480) × 50 = ₹ 1,500 (F)
B = (340 – 320) × 60 = ₹ 1,200 (F)
MYV = ₹ 2,700 (F)

OR Material Yield Variance (MYV)

= (Actual yield – St. yield) × St. cost per unit of output

$$= \left[680 - \left(\frac{680}{850} \times 800\right)\right] \times \left(\frac{45{,}900}{680}\right) = ₹\ 2{,}700\ (F)$$

I/P – loss – O/P
125 – 25 (B/F) = 100
1/5 1/4

Question – 2

SK Ltd. manufactures SK by mixing three raw materials. For each batch of 100 kg of SK, 125 kg of raw material are used. In June 60 batches are prepared to produce an output of 5600 kg of SK. The standard and actual particulars for June are as follows:

Raw materials	Standard		Actual		Quantity of raw material purchased
	Mix %	Price per kg (₹)	Mix %	Price per kg (₹)	
X	50	20	60	21	5000
Y	30	10	20	8	2000
Z	20	5	20	6	1200

Calculate all variances.

Solution

Basic Calculation

Particulars	Standard			Actual			Revised Std. Quantity
	Quantity	Rate	Amount	Quantity	Rate	Amount	
Material X	7,000 × 50% = 3,500	20	70,000	7,500 × 60% = 4,500	21	94,500	7,500 × 50% = 3,750
Material Y	7,000 × 30% = 2,100	10	21,000	7,500 × 20% = 1,500	8	12,000	7,500 × 30% = 2,250
Material Z	7,000 × 20% = 1,400	5	7,000	7,500 × 20% = 1,500	6	9,000	7,500 × 20% = 1,500
Input	7,000		98,000	60×125 = 7,500		1,15,500	7,500
(-) Loss	5,600 × (1/4) = 1,400			1,900			
Output	5,600			5,600			

Calculation of Variances

1. Material Cost Variance = Standard Cost – Actual cost

X = 70,000 – 94,500	= ₹	24,500 (A)
Y = 21,000 – 12,000	= ₹	9,000 (F)
Z = 7,000 – 9,000	= ₹	2,000 (A)
MCV	= ₹	17,500 (A)

Default

2. Material Price Variance = (SP – AP) × AQ → Consumed

	X = (20 – 21) × 4,500	= ₹ 4,500 (A)
	Y = (10 – 8) × 1,500	= ₹ 3,000 (F)
	Z = (5 – 6) × 1,500	= ₹ 1,500 (A)
	MPV	= ₹ 3,000 (A)

Material Price Variance = (SP – AP) × AQ Purch.

(On purchase qty.)	X = (20 – 21) × 5,000	= ₹ 5,000 (A)
	Y = (10 – 8) × 2,000	= ₹ 4,000 (F)
	Z = (5 – 6) × 1,200	= ₹ 1,200 (A)
	MPV	= ₹ 2,200 (A)

3. Material Usage (or Quantity) Variance = (SQ – AQ) × SP

X = (3,500 – 4,500) × 20	= ₹ 20,000 (A)
Y = (2,100 – 1,500) × 10	= ₹ 6,000 (F)
Z = (1,400 – 1,500) × 5	= ₹ 500 (A)
MUV	= ₹ 14,500 (A)

4. Material Mix Variance = (RSQ - AQ) × SP

X = (3,750 – 4,500) × 20	= ₹ 15,000 (A)
Y = (2,250 – 1,500) × 10	= ₹ 7,500 (F)
Z = (1,500 – 1,500) × 5	= ₹ NIL
MMV	= ₹ 7,500 (A)

5. Material Yield Variance = (SQ - RSQ) × SP

X = (3,500 – 3,750) × 20	= ₹ 5,000 (A)
Y = (2,100 – 2,250) × 10	= ₹ 1,500 (A)
Z = (1,400 – 1,500) × 5	= ₹ 500 (A)
MYV	= ₹ 7,000 (A)

OR Material Yield Variance (MYV)

= (Actual yield – St. yield) × St. cost per unit of output

$$= \left[5{,}600 - \left(\frac{100}{125} \times 7{,}500\right)\right] \times \left(\frac{98{,}000}{5{,}600}\right) = ₹\ 7{,}000\ (A)$$

Question – 3

Following data is extracted from the books of SK Ltd. for the month of May, 2021:

(i) Estimation-

Particulars	Quantity (kg)	Price(₹)	Amount (₹)
Material – A	800	?	-
Material – B	600	30.00	18,000
			-

Normal loss was expected to be 10% of total input materials.

(ii) Actuals-

1,480 kg of output produced

Particulars	Quantity (kg)	Price(₹)	Amount (₹)
Material – A	900	?	-
Material – B	?	32.50	-
			59,825

(iii) Other information-

Material cost variance = ₹ 3,625 (F) $= SC - AC \Rightarrow 3625 = SC - 59825 \Rightarrow SC = 63450$

Material price variance = ₹ 175 (F)

$MUV = (SQ - AQ) \times SP$

$3625 - 175 = (940 - 900)(45) + (705 - AQ_B)(30)$

$AQ_B = 650$

You are required to calculate:

(i) Standard price of Material-A

(ii) Actual quantity of Material–B

(iii) Actual price of Material-A

(iv) Revised standard quantity of Material-A and Material-B; and

(v) Material Mix Variance

Solution

(i) Material cost variance = Standard cost – Actual cost

3,625 (F) = Standard cost – 59,825

3,625 = Standard cost – 59,825

Standard cost = 63,450

Total standard input required for actual output = $\frac{1,480}{90\%}$ = 1,645 kg

Standard quantity of material A = $\frac{800}{(800+600)} \times 1,645 = 940$ kg

Standard quantity of material B = $\frac{600}{(800+600)} \times 1,645 = 705$ kg

Standard cost of Material A + Standard cost of Material B = 63,450

$(SQ_A \times SP_A) + (SQ_B \times SP_B) = 63,450$

$(940 \times SP_A) + (705 \times 30) = 63,450$

$SP_A = \frac{42,300}{940}$

Standard price of material A = ₹ 45

(ii) Material price variance = (AQ × SP) – (AQ × AP)

175 (F) = (AQ × SP) – 59,825

175 = (AQ × SP) – 59,825

AQ × SP = 60,000

$(AQ_A \times SP_A) + (AQ_B \times SP_B) = 60,000$

$(900 \times 45) + (AQ_B \times 30) = 60,000$

$AQ_B = \frac{19,500}{30}$

Actual quantity of material B = 650 kg

(iii) Given, AQ × AP = 59,825

$(AQ_A \times AP_A) + (AQ_B \times AP_B) = 59,825$

$(900 \times AP_A) + (650 \times 32.50) = 59,825$

$AP_A = \frac{38,700}{900}$

Actual price of material A = ₹ 43

(iv) Total actual input quantity = 900 + 650 = 1,550 kg

Revised standard quantity of material A = $\frac{800}{(800+600)} \times 1,550$ = 886 kg

Revised standard quantity of material A = $\frac{600}{(800+600)} \times 1,550$ = 664 kg

(v) Material Mix Variance = (RSQ – AQ) × SP

Material A	= (886 – 900) × 45	= ₹ 630 (A)
Material B	= (664 – 650) × 30	= ₹ 420 (F)
		₹ 210 (A)

11. Total Hours paid

- Total hour paid
  - Idle Hours
    - Normal idle hour → Known in Advance → Burden Pass to consumer
    - Abnormal idle hours → After the loss → Burden pass to owner
  - Hours worked

12. Labour Variances

LCV = Std. Cost − Actual Cost

LRV
$= (SR - AR) \times A.\text{Hrs. Paid}$

Idle time Var.
$= \text{Idle Hrs.} \times SR$
[① Always Adverse
② Ab. loss]

LEV
$= (SH - A.\text{Hrs. work}) \times SR$

LMV
$= (RSH - AH_{\text{work}}) \times SR$

LYV
$= (SH - RSH) \times SR$
or
$= (AY - SY) \times SC \text{ P.u. of O/P}$

RSH on basis of A.H. work

13. Labour Cost Variance (LCV)

LCV = Standard cost – Actual cost

14. Labour Rate Variance (LRV)

LRV = (SR – AR) × Actual hours paid

15. Labour Efficiency Variance (LEV)

LEV = (SH – Actual hours worked) × SR

16. Labour Idle Time Variance

Idle time variance = Idle hours × SR

17. Labour Mix Variance (LMV)

LMV = (RSH – AH worked) × SR

SR = Standard rate
AR = Actual rate
SH = Standard hours of labour
AH = Actual hours of labour
RSH = Revised Standard Hours
= Actual working hours in standard ratio
AY = Actual Yield or Output
SY = Standard Yield or Output
$= \frac{\textit{Standard Output}}{\textit{Standard Input}} \times \textit{Actual Input}$

18. Labour Yield Variance (LYV)

LYV = (SH – RSH) × SR

LYV = (AY – SY) × Standard cost per unit of output

Question – 4	

A gang of workers normally consists of 30 skilled workers, 15 semi-skilled workers and 10 unskilled workers. They are paid at standard rate per hour as under:

Skilled ₹ 70

Semi-skilled ₹ 65

Unskilled ₹ 50

In a normal working week of 40 hours, the gang is expected to produce 2,000 units of output. During the week ended 31st March, 2019, the gang consisted of 40 skilled, 10 semi-skilled and 5 unskilled workers. The acutal wages paid were at the rate of ₹ 75, ₹ 60 and ₹ 52 per hour respectively. Four hours were lost due to machine breakdown and 1,600 units were produced.

Calculate the following variances showing clearly adverse (A) or favorable (F)

(i) Labour Cost Variance

(ii) Labour Rate Variance

(iii) Labour Efficiency Variance

(iv) Labour Mix Variance

(v) Labour Idle Time variance

Solution

Basic Calculation

Particulars	Standard (1,600 units)			Actual (1,600 units)			Revised Std. Qty.
	Quantity	Rate	Amount	Quantity	Rate	Amount	
Skilled	$\frac{40\times30}{2,000}\times1600 =$ 960	70	67,200	40×40 = 1,600	75	1,20,000	$\frac{960}{1,760}\times1980$ = 1,080
Semi-skilled	$\frac{40\times15}{2,000}\times1600 =$ 480	65	31,200	40×10 = 400	60	24,000	$\frac{480}{1,760}\times1980$ = 540
Unskilled	$\frac{40\times10}{2,000}\times1600 =$ 320	50	16,000	40×5 = 200	52	10,400	$\frac{320}{1,760}\times1980$ = 360
Total	1,760		1,14,400	2,200		1,54,400	1,980

Particulars	Hours Paid	Idle Hours	Hours Worked
Skilled	40 × 40 =1,600	40 × 4 = 160	1,600 – 160 = 1,440
Semi-skilled	40× 10 = 400	10 × 4 = 40	400 – 40 = 360
Unskilled	40× 5 = 200	5 × 4 = 20	200 – 20 = 180
Total	22,00	220	1,980

Calculation of Variances

(i) Labour Cost Variance = Standard Cost – Actual cost

Skilled	= 67,200 – 1,20,000	= ₹ 52,800 (A)
Semi-skilled	= 31,200 – 24,000	= ₹ 7,200 (F)
Unskilled	= 16,000 – 10,400	= ₹ 5,600 (F)
	LCV	= ₹ 40,000 (A)

(ii) Labour Rate Variance = (SR – AR) × AH paid

Skilled	= (70 – 75) × 1,600	= ₹ 8,000 (A)
Semi-Skilled	= (65 – 60) × 400	= ₹ 2,000 (F)
Unskilled	= (50 – 52) × 200	= ₹ 400 (A)
	LRV	= ₹ 6,400 (A)

(iii) Labour Efficiency Variance = (SH – AH worked) × SR

Skilled	= (960 – 1,440) × 70	= ₹ 33,600 (A)
Semi-Skilled	= (480 – 360) × 65	= ₹ 7,800 (F)
Unskilled	= (320 – 180) × 50	= ₹ 7,000 (F)
	LEV	= ₹ 18,800 (A)

(iv) Labour Mix Variance = (RSH – AH worked) × SR

Skilled	= (1,080 – 1,440) × 70	= ₹ 25,200 (A)
Semi-Skilled	= (540 –360) × 65	= ₹ 11,700 (F)
Unskilled	= (360 – 180) × 50	= ₹ 9,000 (F)
	LMV	= ₹ 4,500 (A)

(v) Idle Time Variance = Idle Hours × SR

Skilled	= 160 × 70	= ₹ 11,200 (A)
Semi-Skilled	= 40 × 65	= ₹ 2,600 (A)
Unskilled	= 20 × 50	= ₹ 1,000 (A)
	Idle time variance	= ₹ 14,800 (A)

Question – 5

The standard output of product 'DJ' is 25 units per hour in manufacturing department of a company employing 100 workers. In a 40 hours week, the department produced 960 units of product 'DJ' despite 5% of the time paid was lost due to an abnormal reason. The hourly wage rates actually paid were ₹ 6.20, ₹ 6.00 and ₹ 5.70 respectively to group 'A' consisting 10 workers, Group 'B' consisting 30 workers and Group 'C' consisting 60 workers. The standard wage rate per labour is same for all the workers. Labour Efficiency Variance is given ₹ 240 (F).

You are required to calculate:

(i) Total Labour Cost Variance
(ii) Total Labour rate Variance
(iii) Total Labour Gang Variance Mix
(iv) Total Labour Yield Variance, and
(v) Total Labour Idle Time Variance

$240(F) = (SH - AH_{work}) \times SR$

$240 = (3840 - 3800) \times SR$

$240 = 40 \times SR$

$SR = \frac{240}{40} = 6$

Solution

Labour Efficiency Variance = (SH – AH worked) × SR

$$240\ (F) = \left[\left(960 \times \frac{100}{25}\right) - \{(10 + 30 + 60) \times (40 - 5\%)\}\right] \times SR$$

240 = (3,840 – 3,800) × SR

SR = ₹ 6

$= \left[960 - \left(\frac{960}{3840} \times 3800\right)\right]\left(\frac{23040}{960}\right)$

Particulars	Standard (960 units)		
	Quantity	Rate	Amount
Labour	$960 \times \frac{100}{25} = 3{,}840$	6	23,040

Actual data (960 units)					
No. of workers	Hours paid	Wage rate	Wages	Idle hours	Hours worked
10	10×40 = 400	6.20	2,480	400×5% = 20	400 – 20 = 380
30	30×40 = 1,200	6	7,200	1,200×5% = 60	1,200 – 60 = 1,140
60	60×40 = 2,400	5.70	13,480	2,400×5% = 120	2,400 – 120 = 2,280
Total	4,000		23,360	200	3,800

Calculation of Variances

(i) Labour Cost Variance = Standard Cost – Actual cost
= 23,040 – 23,360 = ₹ 320 (A)

(ii) Labour Rate Variance = (SR – AR) × AH paid
= [(6 – 6.20) ×400] + [(6 – 6)×1,200] + [(6 - 5.70)×2,400] = ₹ 640 (F)

(iii) Labour Gang Variance = (RSH – AH worked) × SR
= (3,800 – 3,800) × 6 = $\boxed{\text{Nil}}$

(iv) Labour Yield Variance = (Actual yield – St. yield) × St. cost per unit of output
$= \left[960 - \left(\frac{960}{3{,}840} \times 3{,}800\right)\right] \times \left(\frac{23{,}040}{960}\right) =$ ₹ 240 (F)

(v) Idle Time Variance = Idle Hours × SR
= 200 × 6 = ₹ 1,200 (A)

Question – 6

SK Ltd. had prepared the following estimation for the month of January:

	Quantity	Rate (₹)	Amount (₹)
Material – A	800 kg	90.00	72,000
Material – B	600 kg	60.00	36,000
Skilled Labour	1,000 hours	75.00	75,000
Unskilled Labour	800 hours	44.00	35,200

Normal loss was expected to be 10% of total input materials and an idle labour time of 5% of expected labour hours was also estimated. → Known in adv. – NR

At the end of the month the following information has been collected from the cost accounting department:

The company has produced 1,480 kg finished product by using the followings:

	Quantity	Rate (₹)	Amount (₹)
Material – A	900 kg	86.00	77,400
Material – B	650 kg	65.00	42,250
Skilled Labour	1,200 hours	71.00	85,200
Unskilled Labour	860 hours	46.00	39,560

You are required to calculate:

(a) Material cost variance
(b) Material price variance
(c) Material mix variance
(d) Material yield variance
(e) Labour cost variance
(f) Labour efficiency variance
(g) Labour yield variance

Solution

Basic Calculation for material

Particulars	Standard			Actual			Revised Std. Qty.
	Quantity	Rate	Amount	Quantity	Rate	Amount	
Material A	$\frac{8}{14}\times1,644 = 939$	90	84,510	900	86	77,400	$\frac{8}{14}\times1,550=886$
Material B	$\frac{6}{14}\times1,644 = 705$	60	42,300	650	65	42,250	$\frac{6}{14}\times1,550=664$
Input	1,480 ÷ 90% = 1,644		1,26,810	1,550		1,19,650	1,550
(-) Loss	164			70			
Output	1,480			1,480			

Basic Calculation for labour

Particulars	Standard			Actual			Revised Std. Qty.
	Quantity	Rate	Amount	Quantity	Rate	Amount	
Skilled	$\frac{1,000}{1,800}\times 2,008 = 1,115$	75	83,625	1,200	71	85,200	$\frac{1,115}{2,008}\times 2,060 = 1,144$
Unskilled	$\frac{800}{1,800}\times 2,008 = 893$	44	38,852	860	46	39,560	$\frac{893}{2,008}\times 2,060 = 916$
Total	$\frac{1800\times 0.95\times 1,480}{1,400\times 0.90} = 2,008$		1,22,477	2,060		1,24,760	2,060

Calculation of Variances

(a) Material Cost Variance = Standard Cost – Actual cost

A = 84,510 – 77,400 = ₹ 7,110 (F)
B = 42,300 – 42,250 = ₹ 50 (F)
MCV = ₹ 7,160 (F)

(b) Material Price Variance = (SP – AP) × AQ

A = (90 – 86) × 900 = ₹ 3,600 (F)
B = (60 – 65) × 650 = ₹ 3,250 (A)
MPV = ₹ 350 (F)

(c) Material Mix Variance = (RSQ - AQ) × SP

A = (886 – 900) × 90 = ₹ 1,260 (A)
B = (664 – 650) × 60 = ₹ 840 (F)
MMV = ₹ 420 (A)

(d) Material Yield Variance = (SQ - RSQ) × SP

A = (939 – 886) × 90 = ₹ 4,770 (F)
B = (705 – 664) × 60 = ₹ 2,460 (F)
MYV = ₹ 7,230 (F)

OR Material Yield Variance (MYV)

= (Actual yield – St. yield) × St. cost per unit of output

$= \left[1,480 - \left(\frac{1,480}{1,644} \times 1,550\right)\right] \times \left(\frac{1,26,810}{1,480}\right) =$ ₹ 7,251 (F)

(e) Labour Cost Variance = Standard Cost – Actual cost

Skilled = 83,625 – 85,200 = ₹ 1,575 (A)
Unskilled = 38,852 – 39,560 = ₹ 708 (A)
LCV = ₹ 2,283 (A)

(f) Labour Efficiency Variance = (SH – AH worked) × SR

Skilled = (1,115 – 1,200) × 75 = ₹ 6,376 (A)
Unskilled = (893 – 860) × 44 = ₹ 1,452 (F)
LEV = ₹ 4,924 (A)

(g) Labour Yield Variance = (SH - RSH) × SR

Skilled	= (1,115 – 1,144) × 75	= ₹	2,176 (A)
Unskilled	= (893 – 916) × 44	= ₹	1,012 (A)
	LYV	= ₹	3,188 (A)

OR Labour Yield Variance (LYV)

= (Actual yield – St. yield) × St. cost per unit of output

$$= \left[1{,}480 - \left(\frac{1{,}480}{2{,}008} \times 2{,}060\right)\right] \times \left(\frac{1{,}22{,}477}{1{,}480}\right) = ₹\ 3{,}171\ (A)$$

on Basis of A. Hrs. work

19. Variable OHs Variance

V. OH Cost Var. = Rec. OH – Actual OH

V. OH Expd. Var. = (RR – AR) × A. Hrs. work

V. OH Efficiency Var. = (SH – A. Hrs. work) × RR

20. Variable OH Cost Variance (VOCV)

VOCV = Recovered OHs – Actual OHs

21. Variable OH Expenditure Variance (VOEV)

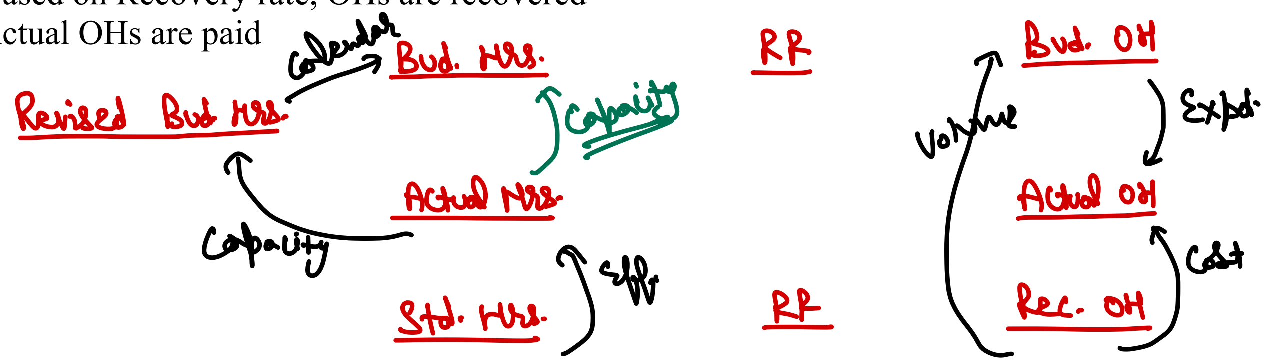
VOEV = (Recover Rate – Actual Rate) × Actual hours worked

22. Variable OH Efficiency Variance (VOEFV)

VOEFV = (SH – Actual hours worked) × Recovery Rate

23. Fixed OHs Variance

- This is based on hours paid
- Fixed OHs are estimated in advance
- Based on estimated OHs, recovery rate is computed
- Based on Recovery rate, OHs are recovered
- Actual OHs are paid



24. Fixed OH Cost Variance (FOCV)

FOCV = Recovered OHs – Actual OHs

25. Fixed OH Expenditure Variance (FOEV)

FOEV = Budgeted OHs – Actual OHs

26. Fixed OH Volume Variance (FOVV)

FOVV = Recovered OHs – Budgeted OHs

27. Fixed OH Efficiency Variance (FOEFV)

FOEFV = (Standard hours – Actual Hours) × Recovery Rate

28. Fixed OH Capacity Variance (FOCPV)

FOCPV = (Actual Hours – Revised budgeted hours) × Recovery Rate

29. Fixed OH Calendar Variance (FOCLV)

FOCLV = (Revised budgeted hours – Budgeted hours) × Recovery Rate

Question – 7

A company operates a standard costing system and showed the following data for the month of March:

	Actual	Budgeted
No. of working days	22	20
Man-hours	4,300	4,000
Overhead rate per hour	-	₹ 0.50
Hours per unit of output	-	10
Fixed overhead incurred	₹ 1,800	-
No. of units produced	425	-

Calculate:

(a) Overhead cost variance
(b) Budget variance
(c) Volume variance
(d) Capacity variance
(e) Calendar variance
(f) Efficiency variance

Rev. BH (22)
$\frac{4000}{20} \times 22 = 4400$

Cal.

B. Hrs (20)
4000

A. Hrs. (22)
4300

Cap.

S. Hrs.
$10 \times 425 = 4250$

Eff.

RR
0.50

RR
0.50

B.OH
2000

Expd.

A.OH
1800

Vol.

R.OH
2125

Cost

Solution

Basic Calculations:

	Budgeted Hours 4,000	**Recovery Rate** 0.50	**Budgeted Overheads** 4,000×0.50 = 2,000
Revised Budgeted Hours $\frac{4,000}{20} \times 22 = 4,400$	**Actual Hours** 4,300		**Actual Overheads** 1,800
	Standard Hours 10 × 425 = 4,250	**Recovery Rate** 0.50	**Recovered Overheads** 2,125

Calculation of Variances

(a) F. O. Cost Variance = Recovered overhead – Actual overhead
= 2,125 – 1,800 = ₹ 325 (F)

(b) Budget Variance = Budgeted overhead – Actual overhead
= 2,000 – 1,800 = ₹ 200 (F)

(c) Volume Variance = Recovered overhead – Budgeted overhead
= 2,125 – 2,000 = ₹ 125 (F)

(d) Capacity Variance = (Actual Hrs. – Revised Budgeted Hrs.) × Recovery Rate
= (4,300 – 4,400) × 0.50 = ₹ 50 (A)

(e) Calendar Variance = (Revised Budgeted Hrs. – Budgeted Hours) × Recovery Rate
= (4,400 – 4,000) × 0.50 = ₹ 200 (F)

(f) Efficiency Variance = (Std. Hrs. – Actual Hrs.) × Recovery Rate
= (4,250 – 4,300) × 0.50 = ₹ 25 (A)

Question – 8

A manufacturing concern has provided following information related to fixed overheads:

	Standard	Actual
Output in a month	5,000 units	4,800 units
Working days in a month	25 days	23 days
Fixed overheads	₹ 5,00,000	₹ 4,90,000

Compute:

(i) Fixed overhead variance
(ii) Fixed overhead expenditure variance
(iii) Fixed overhead volume variance
(iv) Fixed overhead efficiency variance

B.Days 25
A.Days 23
S.Days
$\frac{25}{5000} \times 4800 = 24$
Eff.

RR $\frac{5L}{25} = 20000$
RR 20000

B.OH 5L
A.OH 4.90L
R.OH 4.80L
Expd.
Vol.
Cost

Solution

Basic Calculations:

Budgeted Days 25	**Recovery Rate** 5,00,000 ÷ 25 = 20,000	**Budgeted Overheads** 5,00,000
Actual Days 23		**Actual Overheads** 4,90,000
Standard Days $\frac{25}{5,000} \times 4,800 = 24$	**Recovery Rate** 20,000	**Recovered Overheads** 4,80,000

Calculation of Variances

(a) F. O. Cost Variance = Recovered overhead – Actual overhead
= 4,80,000 – 4,90,000 = ₹ 10,00 (A)

(b) Expenditure Variance = Budgeted overhead – Actual overhead
= 5,00,000 – 4,90,000 = ₹ 10,000 (F)

(c) Volume Variance = Recovered overhead – Budgeted overhead
= 4,80,000 – 5,00,000 = ₹ 20,000 (A)

(d) Efficiency Variance = (Std. days – Actual days) × Recovery Rate
= (24 – 23) × 20,000 = ₹ 20,000 (F)

Question – 9	

PQR Alloys Ltd. uses a standard costing system.

Budgeted information for the year:

Budgeted output	84,000 units
Variable factory overhead per unit	₹ 16
Standard time for one unit of output	0.80 machine hour
Fixed factory overheads	₹ 6,72,000

Actual results for the year:

Actual output	87,600 units
Variable overhead efficiency variance	₹ 67,200 (A)
Actual fixed factory overheads	₹ 7,05,000
Actual variable factory overheads	₹ 14,37,000

Required:

Calculate the following variances clearly indicating Adverse (A) or Favorable (F)

(a) Variable factory overhead expenditure variance
(b) Fixed factory overhead expenditure variance
(c) Fixed factory overhead efficiency variance
(d) Fixed factory overhead capacity variance

Part.	Std. Q	R	A	Actual Q	R	A
V.OH	70080	$\frac{16}{0.8} = 20$	1401600	73440	19.567	14.37L

B. Hrs. $84000 \times 0.8 = 67200$ | RR $\frac{6.72L}{0.672L} = 10$ | B.OH 6.72L

A. hrs. 73440 | A.OH 705000

S. Hrs. $87600 \times 0.8 = 70080$ | RR 10 | R.OH 700800

V. OH Eff. Var. = (SH – AH) × RR
– 67200 = (70080 – AH) × 20
AH = 73440

Solution

Basic Calculations

Particulars	Standard			Actual		
	Hrs.	Rate ₹	Amount ₹	Hrs.	Rate ₹	Amount ₹
Variable Expenses	70,080	20	14,01,600	73,440	$\frac{1437,000}{73,440}$ = 19.567	14,37,000

Budgeted Units 84,000×0.8 = 67,200	**Recovery Rate** $\frac{6,72,000}{67,200} = 10$	**Budgeted Overheads** 6,72,000
Actual Units 73,440		**Actual Overheads** 7,05,000
Standard Units 87,600×0.8 = 70,080	**Recovery Rate** 10	**Recovered Overheads** 7,00,800

Standard rate per hour = $\frac{\textit{Variable factory overhead per unit}}{\textit{Standard time for one unit of output}} = \frac{16}{0.8}$ = ₹ 20

Variable OH efficiency variance = (Standard hours – Actual hours) × Standard rate per hour

- 67,200 = [(87,600 × 0.8) – Actual hours] × 20

-67,200 = (70,080 – Actual hours) × 20

Actual hours = 73,440

(a) Variable factory OH expenditure variance = (Standard rate – Actual Rate)×Actual hours
= (20 – 19.567) × 73,440 = ₹ 31,800 (F)

(b) Fixed factory OH expenditure variance = Budgeted OHs – Actual OHs
= 6,72,000 – 7,05,000 = ₹ 33,000 (A)

(c) Fixed factory OH efficiency variance = (Standard hours – Actual hours) × Recovery rate
= (70,080 – 73,440) × 10 = ₹ 33,600 (A)

(d) Fixed factory OH capacity variance = (Actual hours – Budgeted hours) × Recovery rate
= (73,440 – 67,200) × 10 = ₹ 62,400 (F)

Question – 10

ABC Ltd. has furnished the following information regarding the overheads for the month of June 2020:

(i) Fixed overhead cost variance → ₹ 2,800 (Adverse)
(ii) Fixed overhead volume variance → ₹ 2,000 (Adverse) ✓
(iii) Budgeted Hours for June, 2020 → 2,400 hours } RR = 5
(iv) Budgeted Overheads for June, 2020 ✓ → ✓ ₹ 12,000
(v) Actual rate of recovery of overheads → ₹ 8 per hour

From the above given information calculate:

(i) Fixed overhead expenditure variance → 2800 (A) – 2000 (A) = 800 (A)
(ii) Actual overheads incurred → Exp. Var. = B.OH – A.OH
- 800 = 12000 – A.OH
A.OH = 12800 ✓

$\rightarrow \frac{12800}{8} = 1600$

(iii) Actual hours for actual production

(iv) Fixed overhead capacity variance $(AH - BH) \times RR = (1600 - 2400) \times 5 = 4000\,(A)$

(v) Standard hours for actual production $\rightarrow$ Vol. Var. = Rec. − Bud.

$-2000 = (SH \times 5) - 12000 \Rightarrow SH = 2000$

(vi) Fixed overhead efficiency variance

$\hookrightarrow (SH - AH) \times RR = (2000 - 1600) \times 5 = 2000\,(F)$

Solution

Computation of required variances for June 2020:

1. Overheads expenditure variance = Overhead Cost Variance – Overheads Volume variance

= ₹ 2,800(A) - ₹ 2,000(A) = - ₹ 2,800 – (- ₹ 2,000) = - ₹ 800 = ₹ 800 (A)

2. Actual Overheads incurred = Budgeted Overhead - Overhead Expenditure Variance

= ₹ 12,000 - ₹ 800(A) = ₹ 12,000 – (- ₹ 800) = ₹ 12,000 + ₹ 800 = ₹ 12,800

3. Actual hours for actual production = $\frac{Actual\ Overheads\ Incurred}{Actual\ Rate\ of\ Recovery\ Overhead\ Per\ Hour} = \frac{12{,}800}{8}$ = 1,600 hours

4. Overheads Capacity Variance = Std rate of OH rate (Actual hrs for actual production – Budgeted hours)

= ₹ 5 × (1,600 – 2,400) = ₹ 5 × (- 800) = ₹ 4,000 Adverse

* Standard rate of Overhead recovery = $\frac{Budgetary\ Overheads}{Budgeted\ hours} = \frac{12{,}000}{2{,}400\ hours}$ = ₹ 5 per hour

5. Volume Variance = Std. rate of OHs recovery (Standard hours for actual production – Budgeted hours)

or, ₹ 2,000(A) = ₹ 5 [Std. hrs. – 2,400 hours]

or, Std. hrs. – 2,400 hours = $-\frac{2{,}000}{5}$

or, Std. hrs. – 2,400 hours = - 400 hours

or, Std. hrs. = 2,400 hours - 400 hours

Standard hours for actual production = 2,000 hours

6. Fixed overhead efficiency variance = (Std. hours for AO – Actual Hrs.) × SR

= (2,000 – 1,600) × 5 = 2,000 (F)

Or

Fixed overhead efficiency variance = Volume variance – Capacity Variance

= ₹ 2,000(A) - ₹ 4,000(A) = - ₹ 2,000 – (- ₹ 4,000) = - ₹ 2,000 + ₹ 4,000 = ₹ 2,000 (F)

Question – 11	P.N. = $\frac{1{,}20L}{12}$ = 10000

K Ltd. is a manufacturer of a single product A. 8,000 units of the product A has been produced in the month of March 2024. At the beginning of the year a total 1,20,000 units of the product-A has been planned for production. The cost department has provided the following estimates of overheads:

Particulars (Yr.) (Month)	Amount (₹)	RR
Fixed = 12L + (1.80L × 60%) = 13.08L ÷ 12 = 1.09L	12,00,000	$\frac{1.09L}{10000}$ = 10.9
Semi-variable	60% 1,80,000	
Variable = 6L + (1.80L × 40%) = 6.72L ÷ 12 = 56000	40% 6,00,000	$\frac{56000}{10000}$ = 5.6

Semi-variable charges are considered to include 60 per cent expenses of fixed nature and 40 per cent of variable character.

The records of the production department shows that the company could have operated for 20 days but there was a festival holiday during the month.

The actual cost data for the month of March 2024 are as follows:

Particulars	Amount (₹)
Fixed = 110000 + (19200×60%) = 121520 ✓	1,10,000
Semi-variable	19,200
Variable = 48000 + (19200×40%) = 55680	48,000

The cost department of the company is now preparing a cost variance report for managerial information and action. You being an accounts officer of the company are asked to calculate the following information for preparation of the variance report:

Question – 1

What is the amount of variable overhead cost variance for the month of March 2024:

(a) ₹ 10,200 (A)
(b) ₹ 10,400 (A)
(c) ₹ 10,800 (A)
(d) ₹ 10,880 (A) ✓

Rec. OH − A. OH
(5.6×8000) − 55680 = 10880 (A)

Question – 2

What is the amount of fixed overhead volume variance for the month of March:

(a) ₹ 9,000 (F)
(b) ₹ 9,000 (A)
(c) ₹ 21,800 (A) ✓
(d) ₹ 11,000 (A)

Rec. OH − Bud. OH
(10.9×8000) − 1.09L = 21800 (A)

Question – 3

What is the amount of fixed overhead expenditure variance for the month of March:

(a) ₹ 21,520 (A) ✓ 12
(b) ₹ 21,500 (A)
(c) ₹ 21,400 (A)
(d) ₹ 21,480 (A)

Bud. OH − A. OH
1.09L − 121520 = 12520 (A)

Question – 4

What is the amount of fixed overhead calendar variance for the month of March:

(a) ₹ 5,400 (A)
(b) ₹ 5,450 (A) ✓
(c) ₹ 5,480 (A)
(d) ₹ 5,420 (A)

(RBH − BH) × RR
$\left[\left(\frac{10000}{20} \times 19\right) - 10000\right] \times 10.9 = 5450\,(A)$

Question – 5

What is the amount of fixed overhead cost variance for the month of March:

(a) ₹ 43,220 (A)
(b) ₹ 43,300 (A)
(c) ₹ 43,200 (A)
(d) ₹ 43,380 (A)

Rec. - Actual
(10.9 × 8000) — 121520 = 34320 (A)

1	2	3	4	5
D	C	A	B	A

30. Budget Ratios

1. Efficiency Ratio $= \frac{Standard\ hours}{Actual\ hours} \times 100$

2. Activity Ratio $= \frac{Standard\ hours}{Budgeted\ hours} \times 100$ = Eff. × Cap. Ratio

3. Calendar Ratio $= \frac{Actual\ working\ days}{Budgeted\ Working\ Days} \times 100 = \frac{Revised\ budgeted\ hours}{Budgeted\ hours} \times 100$

4. Actual usage of Budgeted Capacity Ratio $= \frac{Actual\ hours}{Budgeted\ hours} \times 100$

5. Standard Capacity Ratio $= \frac{Budgeted\ hours}{Maximum\ possible\ hours\ in\ budget} \times 100$

6. Actual Capacity Usage Ratio $= \frac{Actual\ hours}{Maximum\ possible\ working\ hours} \times 100$

Question – 12	

Following data is available for ABC Ltd.:

Standard working hours	8 hours per day of 5 days per week
Maximum capacity	→ 60 employees
Actual working	→ 50 employees
Actual hours expected to be worked per four week	~~8,000 hours~~
Standard hours expected to be earned per four week	~~9,600 hours~~
Actual hours worked in the four week period	✓ 7,500 hours AH
Standard hours earned in the four week period	✓ 8,800 hours SH

BH = 8 × 5 × 4 × 50 = 8000
Max. Hrs. = 8 × 5 × 4 × 60
= 9600

The related period is of four weeks. Calculate the following Ratios:

(i) Efficiency ratio → 8800 ÷ 7500 = 117.33%
(ii) Activity ratio → 8800 ÷ 8000 = 110%
(iii) Standard capacity usage ratio → 8000 ÷ 9600 = 83.33%
(iv) Actual capacity usage ratio → 7500 ÷ 9600 = 78.125%
(v) Actual usage of Budgeted capacity ratio
→ 7500 ÷ 8000 = 93.75

Solution

Working Notes:

(1) Max. capacity in a budget period = 60 employees × 8 hrs. × 5 days × 4 weeks = 9,600 hrs.

(2) Budgeted hours = 50 employees × 8 hrs. × 5 days × 4 weeks = 8,000 hrs.

(3) Actual hours = 7,500 hrs. (given)

(4) Standard hours for actual output = 8,800 hours

Calculation of ratios:

(i) Efficiency ratio = $\frac{Standard\ Hours}{Actual\ Hours} \times 100 = \frac{8,800}{7,500} \times 100 = 117.33\%$

(ii) Activity ratio = $\frac{Standard\ Hours}{Budgeted\ Hours} \times 100 = \frac{8,800}{8,000} \times 100 = 110\%$

(iii) Standard Capacity Usage ratio = $\frac{Budgeted\ Hours}{Max.\ possible\ hours\ in\ budget\ period} \times 100 = \frac{8,000}{9,600} \times 100 = 83.33\%$

(iv) Actual capacity usage ratio = $\frac{Actual\ Hours}{Max.\ possible\ working\ hours\ in\ period} \times 100 = \frac{7,500}{9,600} \times 100 = 78.125\%$

(v) Actual usage of budgeted capacity ratio = $\frac{Actual\ hours}{Budgeted\ Days} \times 100 = \frac{7,500}{8,000} \times 100 = 93.75\%$